

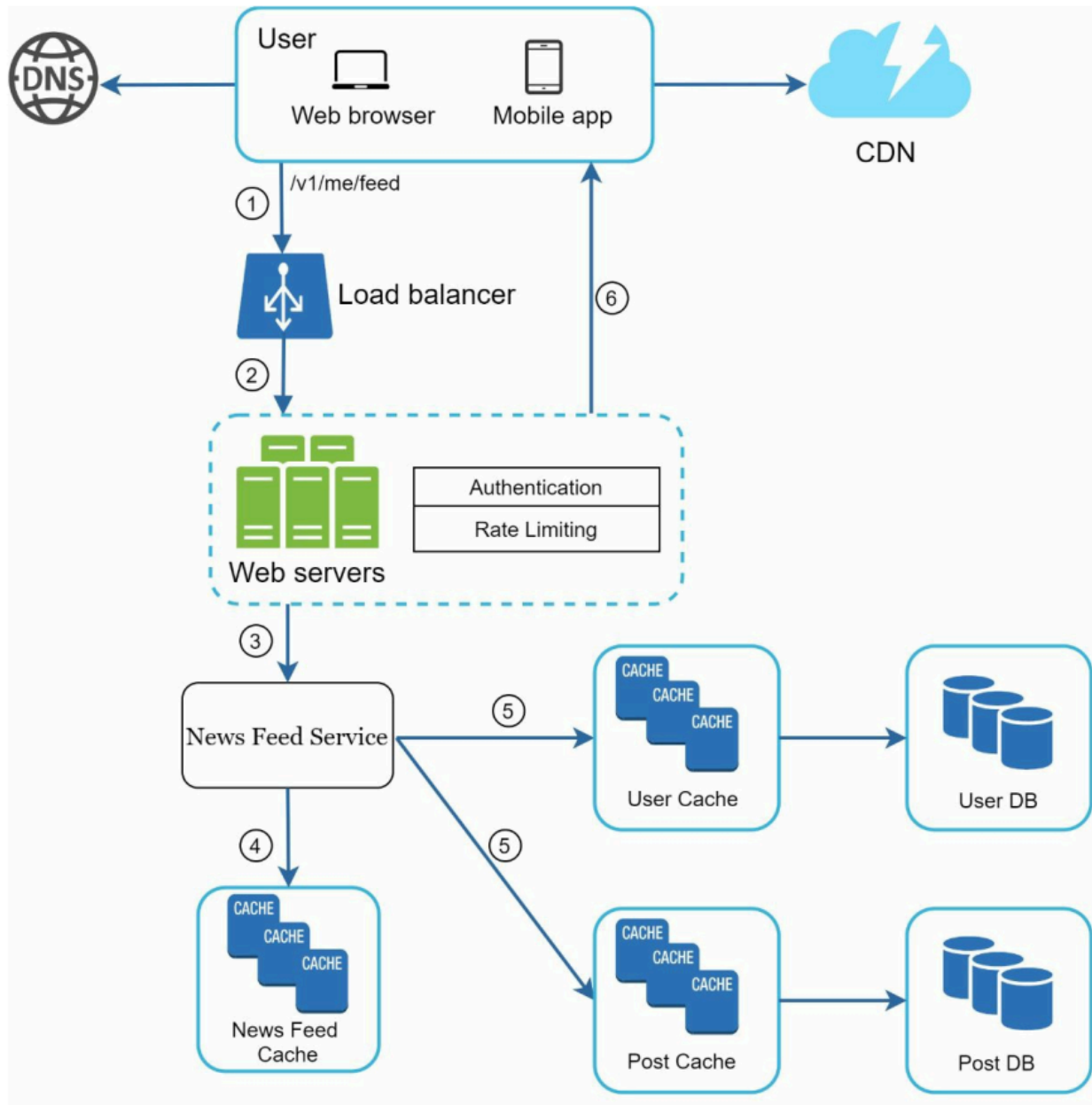
Design a News Feed System - Newsfeed Retrieval Deep Dive

1. Goal of Newsfeed Retrieval

The purpose of the newsfeed retrieval flow is to:

- Fetch posts efficiently
- Return posts in reverse chronological order
- Serve content with very low latency
- Support large-scale traffic efficiently

2. Detailed Retrieval Flow



The following steps occur when a user opens the news feed page.

Step 1: User Requests News Feed

The client sends a request:

None

GET /v1/me/feed

This request asks the system to retrieve the latest news feed posts.

Step 2: Load Balancer Routes Request

- The load balancer distributes incoming traffic
- Requests are routed to available web servers

Purpose:

- Prevent server overload
- Improve scalability and availability

Step 3: Web Servers Call News Feed Service

- Web servers authenticate the request
- The request is forwarded to the News Feed Service

The News Feed Service is responsible for assembling the feed data.

Step 4: Fetch Feed IDs from News Feed Cache

- News Feed Service queries the News Feed Cache
- The cache stores a list of post IDs for each user

Example:

None

User A → [Post101, Post102, Post103]

Using cache greatly reduces database load and improves response time.

Step 5: Hydrate Feed Data

A news feed contains more than just post IDs.

The system fetches additional details such as:

- Username
- Profile picture
- Post text
- Images/videos
- Timestamps
- Like counts
- Comments

To build the complete feed:

- User Cache stores user information
- Post Cache stores post content

The News Feed Service combines all these pieces together.

This process is called:

Feed Hydration

Step 6: Media Content from CDN

Large media files such as:

- Images
- Videos
- GIFs

are stored in a CDN (Content Delivery Network).

Benefits:

- Faster media loading
- Reduced backend server load
- Better global performance

Step 7: Return Final Feed to Client

The fully hydrated feed is returned to the client in JSON format.

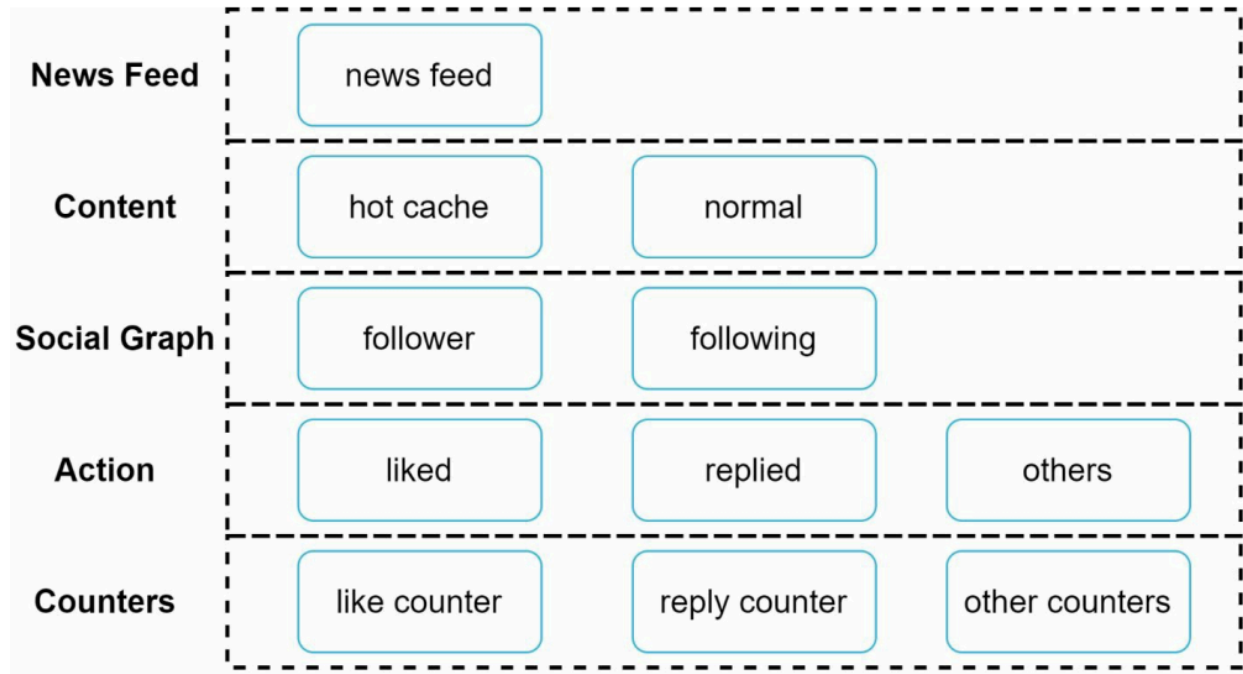
Example response:

JSON

```
{  
  "user": "Sam",  
  "post": "Hello World",  
  "image": "cdn-link",  
  "likes": 120  
}
```

The client application renders the feed on screen.

3. Cache Architecture



Cache is one of the most important parts of a news feed system.

The cache layer is divided into multiple specialized caches.

4. Types of Cache Layers

A. News Feed Cache

Stores:

- Feed IDs for each user

Example:

None

User A → [P1, P2, P3]

Purpose:

- Fast retrieval of feed ordering

B. Content Cache

Stores:

- Actual post data

Includes:

- Text
- Images
- Videos
- Metadata

Popular posts are stored in:

Hot Cache

Benefits:

- Reduces database queries
- Improves feed loading speed

C. Social Graph Cache

Stores:

- Friend relationships
- Followers/following data

Purpose:

- Quickly identify whose posts should appear in feed

D. Action Cache

Stores user actions such as:

- Likes
- Comments
- Shares
- Replies

Purpose:

- Fast interaction updates

E. Counter Cache

Stores counters like:

- Number of likes
- Replies
- Followers
- Following count

Purpose:

- Avoid expensive counting queries repeatedly

5. Why Cache is Critical

Without caching:

- Database becomes overloaded
- Feed retrieval becomes slow
- High latency affects user experience

With caching:

- Feed loads quickly
- System scales to millions of users
- Backend pressure reduces significantly

6. CDN Usage

CDN is heavily used for media delivery.

Benefits include:

- Faster image/video loading
- Reduced bandwidth usage
- Improved global access speed
- Lower latency

7. Important Optimizations

Common optimizations include:

- Hot cache for trending posts
- Pagination for large feeds
- Lazy loading of images/videos
- Compression for responses
- CDN edge caching

8. Key Takeaways

Newsfeed retrieval works by:

1. Fetching feed IDs from cache
2. Hydrating full post/user data
3. Serving media through CDN
4. Returning optimized JSON responses

Multiple cache layers are essential for scalability and low latency.

CDN significantly improves media delivery performance.